USGS Data Download Guide

Overview

This guide covers downloading water level data from USGS using the `dataretrieval` Python package. USGS maintains thousands of stream gages across the United States that record water levels at 15-minute intervals.

Installation
```bash
pip install dataretrieval
```

nwis Module (Recommended)

The NWIS module is reliable and straightforward for accessing gage height data.
```python
from dataretrieval import nwis

# Get instantaneous values (15-min intervals)
df, meta = nwis.get_iv(
sites='<station_id>',
start='<start_date>',
end='<end_date>',
parameterCd='00065'
)

# Get daily values
df, meta = nwis.get_dv(
sites='<station_id>',
start='<start_date>',
end='<end_date>',
parameterCd='00060'
```

)

Get site information
info, meta = nwis.get_info(sites='<station_id>')
```

## Parameter Codes

| Code | Parameter | Unit | Description |
|---|---|---|---|
| `00065` | Gage height | feet | Water level above datum |
| `00060` | Discharge | cfs | Streamflow volume |

## nwis Module Functions

| Function | Description | Data Frequency |
|---|---|---|
| `nwis.get_iv()` | Instantaneous values | ~15 minutes |
| `nwis.get_dv()` | Daily values | Daily |
| `nwis.get_info()` | Site information | N/A |
| `nwis.get_stats()` | Statistical summaries | N/A |
| `nwis.get_peaks()` | Annual peak discharge | Annual |

## Returned DataFrame Structure

The DataFrame has a datetime index and these columns:

| Column | Description |
|---|---|
| `site_no` | Station ID |
| `00065` | Water level value |
| `00065_cd` | Quality code (can ignore) |

## Downloading Multiple Stations
```python
from dataretrieval import nwis

station_ids = ['<id_1>', '<id_2>', '<id_3>']
all_data = {}

for site_id in station_ids:
try:
df, meta = nwis.get_iv(
sites=site_id,
start='<start_date>',
end='<end_date>',
parameterCd='00065'
)
if len(df) > 0:
all_data[site_id] = df

except Exception as e:
print(f"Failed to download {site_id}: {e}")

print(f"Successfully downloaded: {len(all_data)} stations")
```

Extracting the Value Column
```python
# Find the gage height column (excludes quality code column)
gage_col = [c for c in df.columns if '00065' in str(c) and '_cd' not in str(c)]

if gage_col:
water_levels = df[gage_col[0]]
print(water_levels.head())
```

Common Issues

Issue	Cause	Solution
Empty DataFrame	Station has no data for date range	Try different dates or use `get_iv()`
`get_dv()` returns empty	No daily gage height data	Use `get_iv()` and aggregate
Connection error	Network issue	Wrap in try/except, retry
Rate limited	Too many requests	Add delays between requests

Best Practices

- Always wrap API calls in try/except for failed downloads
- Check `len(df) > 0` before processing
- Station IDs are 8-digit strings with leading zeros (e.g., '04119000')
- Use `get_iv()` for gage height, as daily data is often unavailable
- Filter columns to exclude quality code columns (`_cd`)
- Break up large requests into smaller time periods to avoid timeouts
```